

Pepperfry Furniture E-Commerce Database Management System

CHAPTER-1

1. Introduction

Pepperfry is one of India's leading online furniture and home decor shopping platforms. It offers a wide range of products, including sofas, beds, dining tables, wardrobes, office furniture, lighting, and home décor items. Customers can browse products, compare prices, place orders, make secure payments, and track deliveries online. The Pepperfry Database Management System is designed to store and manage customer information, product details, orders, payments, inventory, and delivery records efficiently. This database helps improve business operations, ensures data accuracy, and provides a better shopping experience for customers.

1.1 Problem Statement

Managing a large number of products, customers, orders, and payments manually is difficult and time-consuming. Manual record keeping can lead to errors, duplicate data, and delays in processing orders. Therefore, a well-designed database system is required to organize and manage all business information efficiently. The Pepperfry Database Management System solves these problems by storing data securely and enabling quick access to information.

1.2 Objectives

The main objectives of the Pepperfry Database Management System are as follows:

- To design an efficient database for the Pepperfry e-commerce platform.
- To store customer, product, and order information accurately.
- To manage inventory and product categories effectively.
- To maintain secure payment and delivery records.
- To reduce data redundancy and improve data consistency.
- To support fast data retrieval using SQL queries.

1.3 Motive

The main motive of this project is to develop a reliable database system that simplifies the management of an online furniture store. It helps organize business data, improves operational efficiency, and provides customers with a smooth shopping experience.

1.4 Stakeholders

The following stakeholders play an important role in the successful operation of the Pepperfry e-commerce platform:

- *Customers* – Purchase furniture and home décor products through the Pepperfry website.
- *Admin* – Manages products, users, orders, and overall system operations.
- *Product Managers* – Add, update, and organize product information and categories.
- *Suppliers* – Provide furniture and home décor products to the company.
- *Warehouse Staff* – Manage product inventory, storage, and order packing.
- *Delivery Partners* – Deliver products safely and on time to customers.
- *Finance Department* – Handles payments, refunds, invoices, and financial records.
- *Business Management* – Monitors business performance and makes strategic decisions.

1.5 Business Requirements

The system should support the daily business operations of the Pepperfry e-commerce platform by:

- Maintain customer registration details.
- Store furniture and home décor product information.

- Manage product categories and stock availability.
- Process customer orders and payments.
- Track product delivery status.
- Generate reports on sales and inventory.

1.6 Functional Requirements

The system should provide the following functional features to ensure smooth business operations:

- Customer registration and login.
- Add, update, and delete products.
- Search and filter products.
- Add products to the shopping cart.
- Place and manage orders.
- Process online payments.
- Track order delivery.
- Manage inventory and stock.
- View customer reviews.

1.7 Non-Functional Requirements

The system should satisfy the following non-functional requirements to ensure quality, reliability, and security:

- High data security.
- Fast system performance.
- Reliable database operations.
- Easy-to-use interface.
- Scalability for future growth.

- Data backup and recovery.
- High availability and accuracy.

1.8 Conclusion

The Pepperfry Database Management System provides an efficient solution for managing an online furniture business. It helps organize customer, product, order, payment, and delivery information in a structured manner. The system improves business efficiency, reduces errors, ensures data security, and supports better decision-making. Overall, it provides a reliable and user-friendly platform for managing e-commerce operations.